Reading a research claim

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Reading a research claim before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A question that evidence can answer

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Reading a research claim trace and record the first step you can explain without help.

Use W2: Trace the worked case for Reading a research **claim**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Reading a research claim explanation against the supplied facts before discussing it.

Use the worked case for Reading a research **claim**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Report **claim**s universal speed after median2 seconds versus3 on four local fixtures. Audit scope.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

Explain the mechanism. Apply the supplied Reading a research claim facts and keep your

> reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Reading a research claim facts and keep your reasoning.

A fictional report says method B is better after6/8 supported answers versus baseline5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Reading a research claim facts and keep your reasoning.

A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Reading a research claim facts and keep your reasoning.

Report **claim**s universal speed after median2 seconds versus3 on four local fixtures. Audit scope.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Reading a research claim facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Reading a research claim to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled reading a research **claim evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reading a research **claim**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

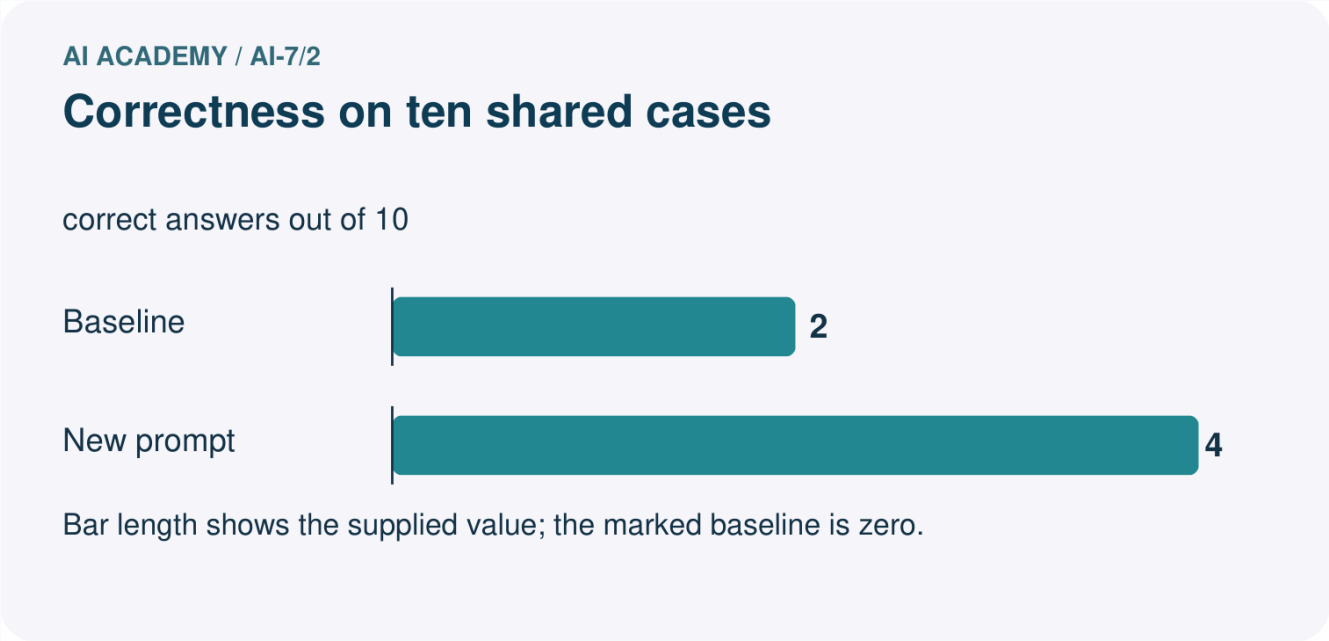
WORKBOOK · 12 OF 15

Pictorial case practice: Reading a research claim

> Read a claim with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two. Explain the labelled step before checking the changed condition.

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

Baseline: 2/10 correct. New prompt: 4/10 on the same ten cases. Scoring is fixed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/2

Reading a research claim

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Find the measured outcome

The table counts correct answers, not learning or satisfaction.



2. Calculate both changes

20% -> 40% is twice the rate and a 20-percentage-point increase.



3. Check claim scope

These ten cases support a descriptive comparison, not universal doubling.

CHANGE ONE THING

The headline changes to "every learner improves twice as much".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? These ten cases support a descriptive comparison, not universal doubling.

1. Check **claim** scope

2. Calculate both changes

3. Find the measured outcome

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check **claim** scope These ten cases support a descriptive comparison, not universal doubling. A research **claim** compresses a method and result.

Unpack the measured variable, denominator and comparison before accepting the headline.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The headline changes to "every learner improves twice as much". Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. That learner **claim** is unsupported by these answer counts.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

Repair a missing building block

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

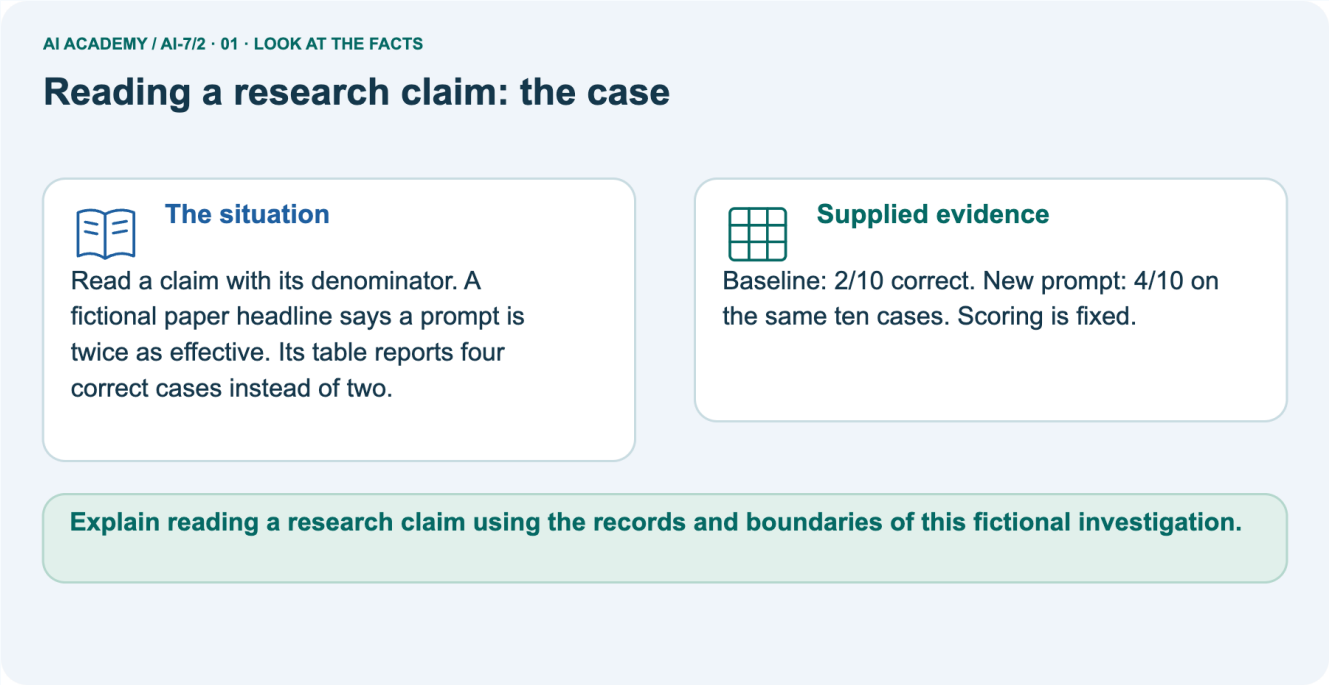
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Reading a research claim picture from its labels, then record a relevant limit.

01. Reading a research claim: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

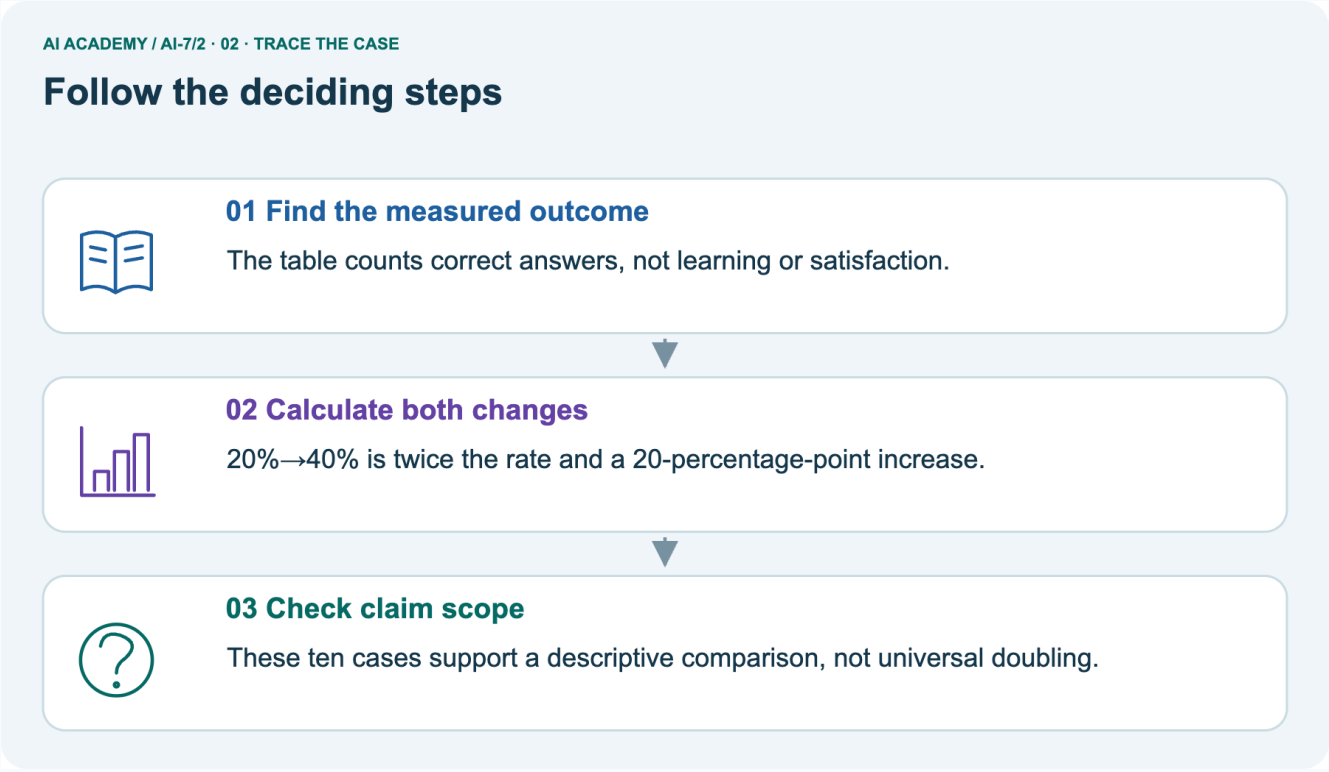
Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two. Supplied facts: Baseline: 2/10 correct.

New prompt: 4/10 on the same ten cases. Scoring is fixed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



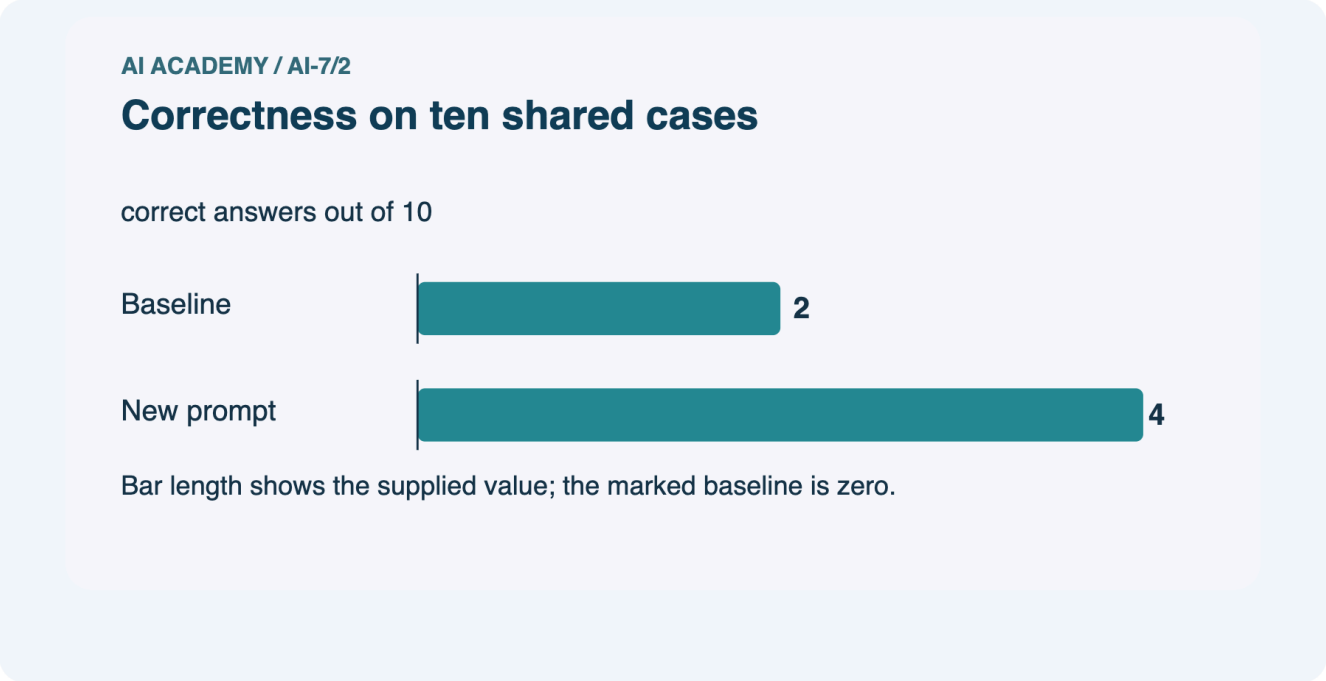
Follow the labelled picture with the matching explanation. Open the vector diagram

Find the measured outcome: The table counts correct answers, not learning or satisfaction. → Calculate both changes: 20%→40% is twice the rate and a 20-percentage-point increase. → Check **claim** scope: These ten cases support a descriptive comparison, not universal doubling.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Correctness on ten shared cases



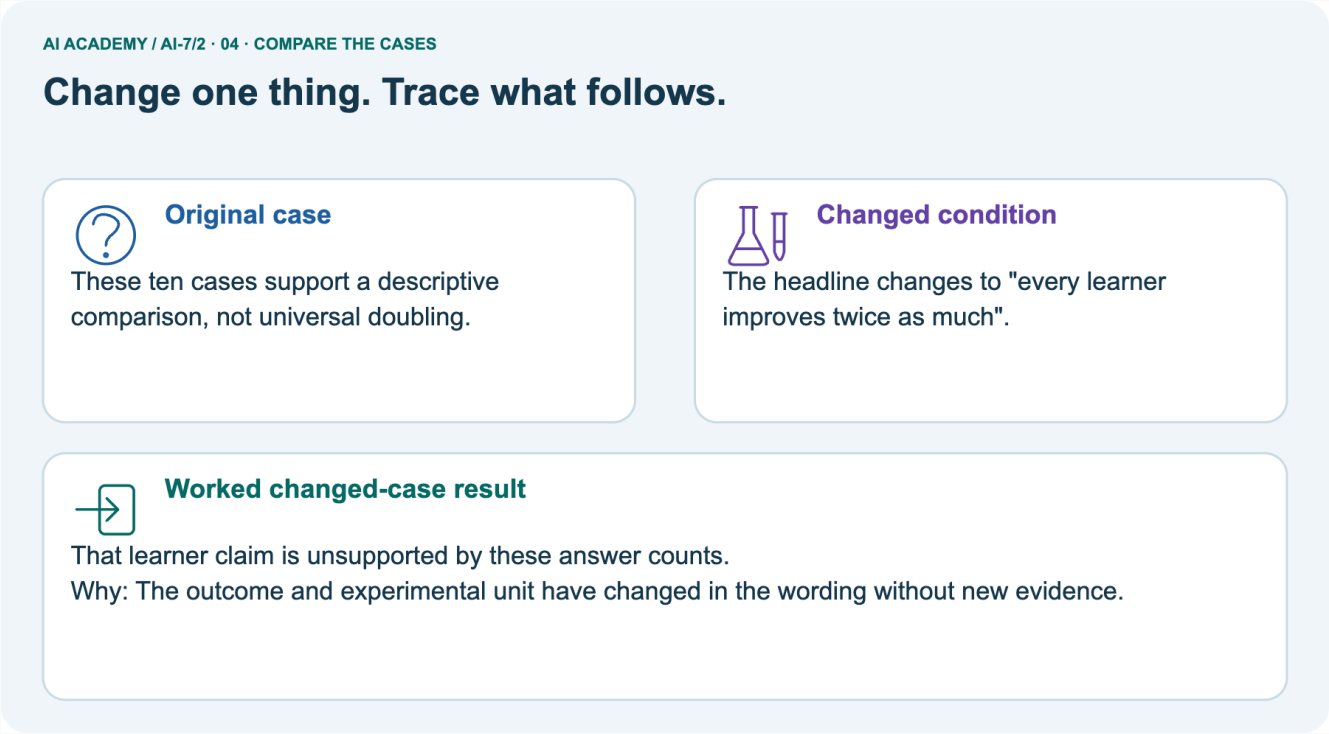
Follow the labelled picture with the matching explanation. Open the vector diagram

Correctness on ten shared cases. Units: correct answers out of 10. Baseline: 2. New prompt: 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

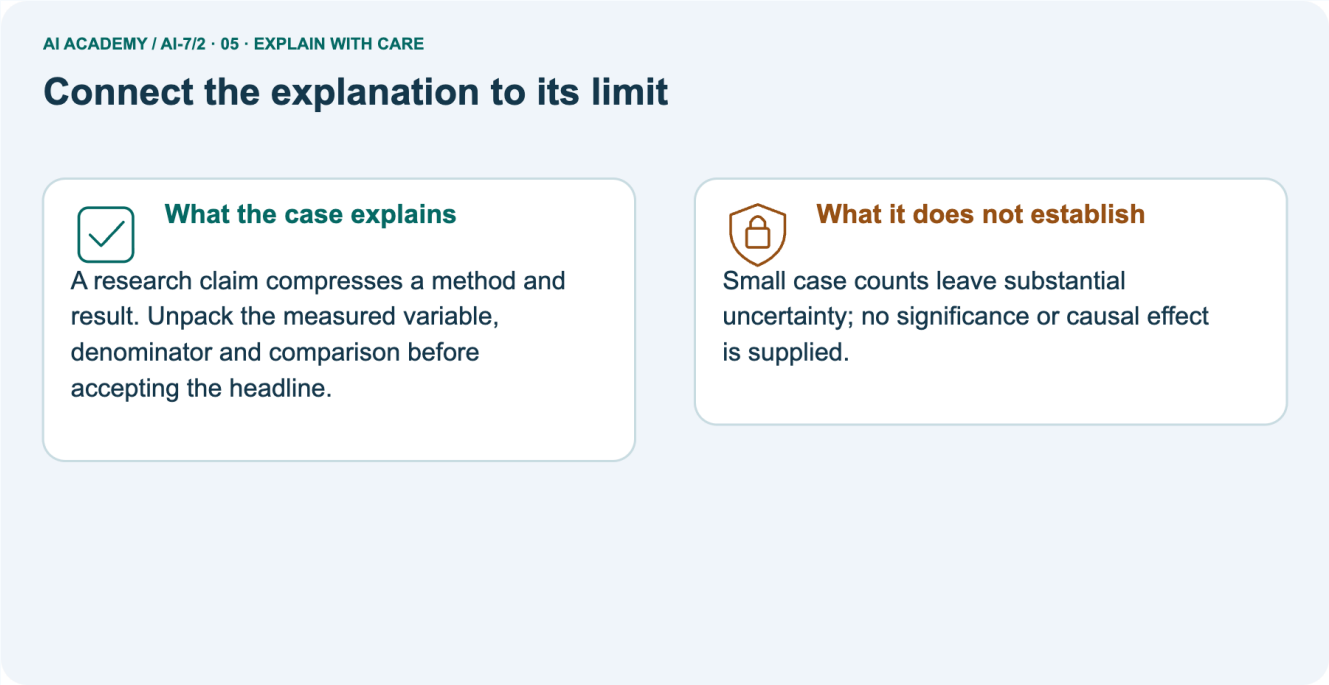
Original case: These ten cases support a descriptive comparison, not universal doubling. Changed condition: The headline changes to "every learner improves twice as much".

Expected result: That learner **claim** is unsupported by these answer counts. Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

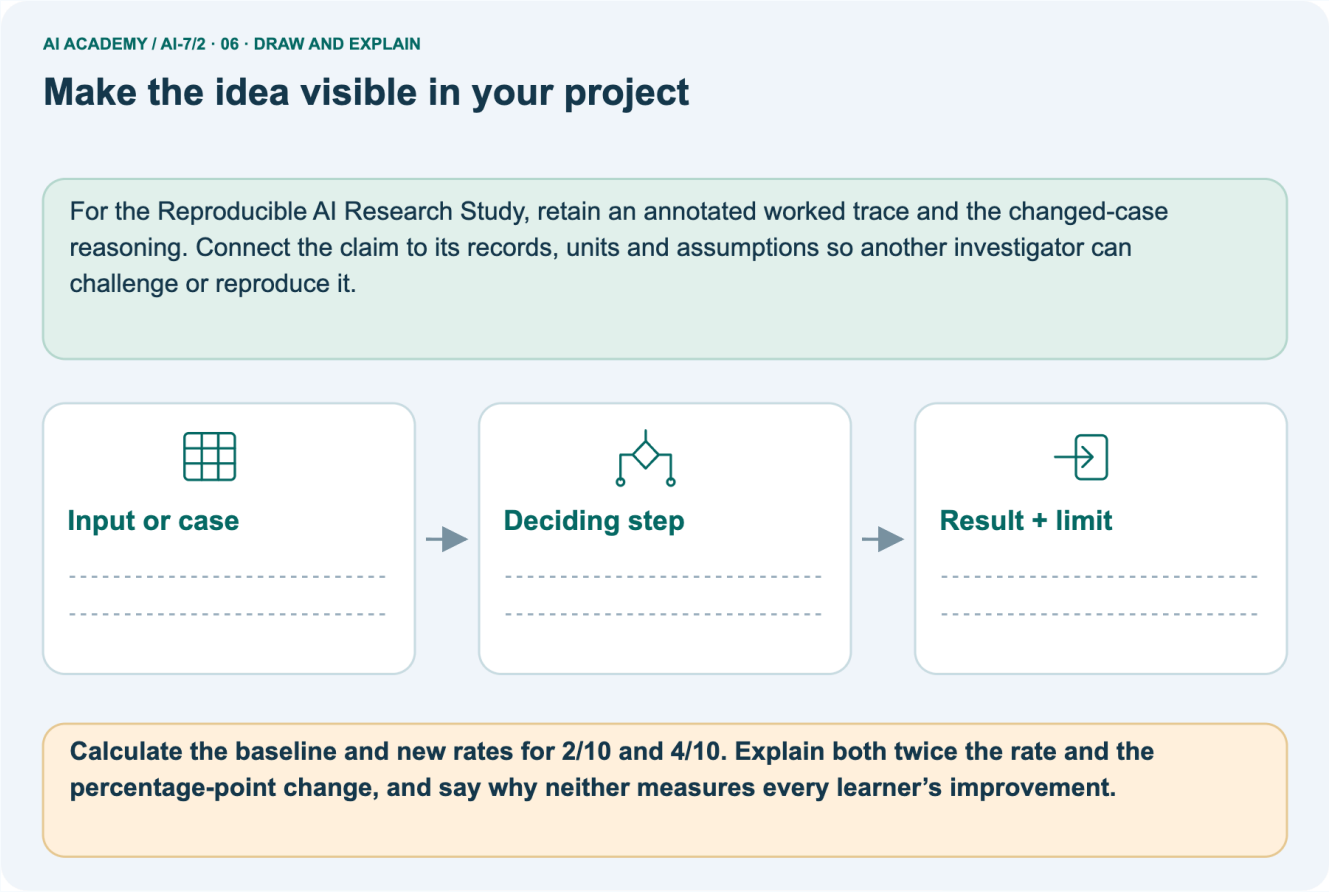
A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Limit: Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

> Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.